

Background & Problem Identification

The proposed research investigated the impact of state required nitrification action or management plans (NAPs) on system nitrification for those practicing chloramination. With more systems switching to chloramination for secondary disinfection, nitrification in the distribution system and losing the disinfectant residual is a growing concern. Nitrate and nitrite levels in the distribution system are an ongoing issue with some states already moving forward with their own regulatory requirements. This project collected data and information on the current states of nitrification action or control plans across the country.

The goal of this research project is to answer questions such as:

1. Would states benefit from mandating a Nitrification Action Plan?
2. What are the potential tradeoffs of such a mandate?

Research Approach

Research OBJECTIVES:

- RO1 - Identify the percent of states that have implemented state enforceable nitrification action/management/control plans**
- RO2 - Evaluate if states with nitrification action/management/control plans see an increase in disinfectant residual and fewer total coliform (TC), and MDBPs (paired utility study).**
- RO3 - Evaluate if, for states that have enacted a nitrification action/management/control plan recently, utilities see a reduction in Nitrite (call for help! future work).**

Dataset 1: Occurrence data

Six-Year Review (SYR) data by PWSID for the following contaminants:

- Free Chlorine
- Total Chlorine
- Positive Total Coliform
- Haloacetic Acids (HAA5)
- Total Trihalomethanes (TTHMs)

Does not report nitrate-nitrite and disinfection type

Dataset 2: to identify disinfection type

UCMR4 data, to get treatment and system information by PWSID. Limited to large systems

Dataset 3: Additional data

Additional State level data/SDWIS violation data:

- Treatment info and disinfection type
- Nitrate & Nitrite
- Free Ammonia
- Temperature
- pH

Limited data available for only Louisiana & Texas

Equation for Intervention Analysis

$$Y_t = \beta_0 + \beta_1 t + \beta_2 D_t + \beta_3 [t - T_I] D_t + \varepsilon_t$$

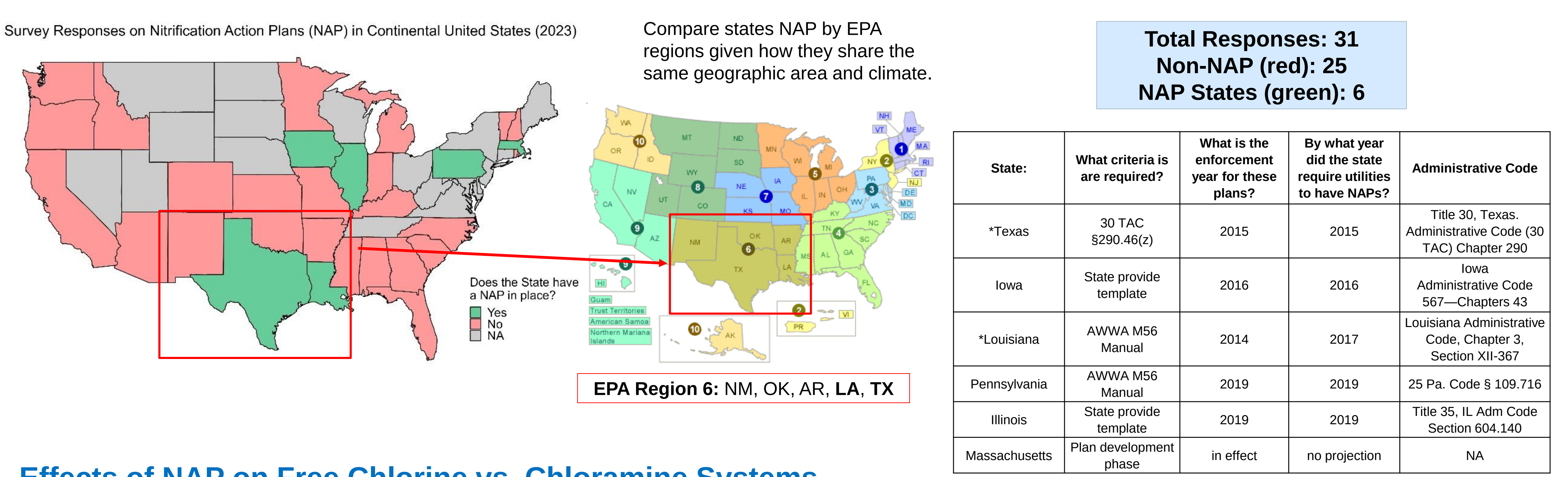
$$Y_t = \text{Intercept} + \text{PrePolicy}t + \text{Policy}D_t + \text{PostPolicy}[t - T_I]D_t + \text{Random Error}$$

Variables:

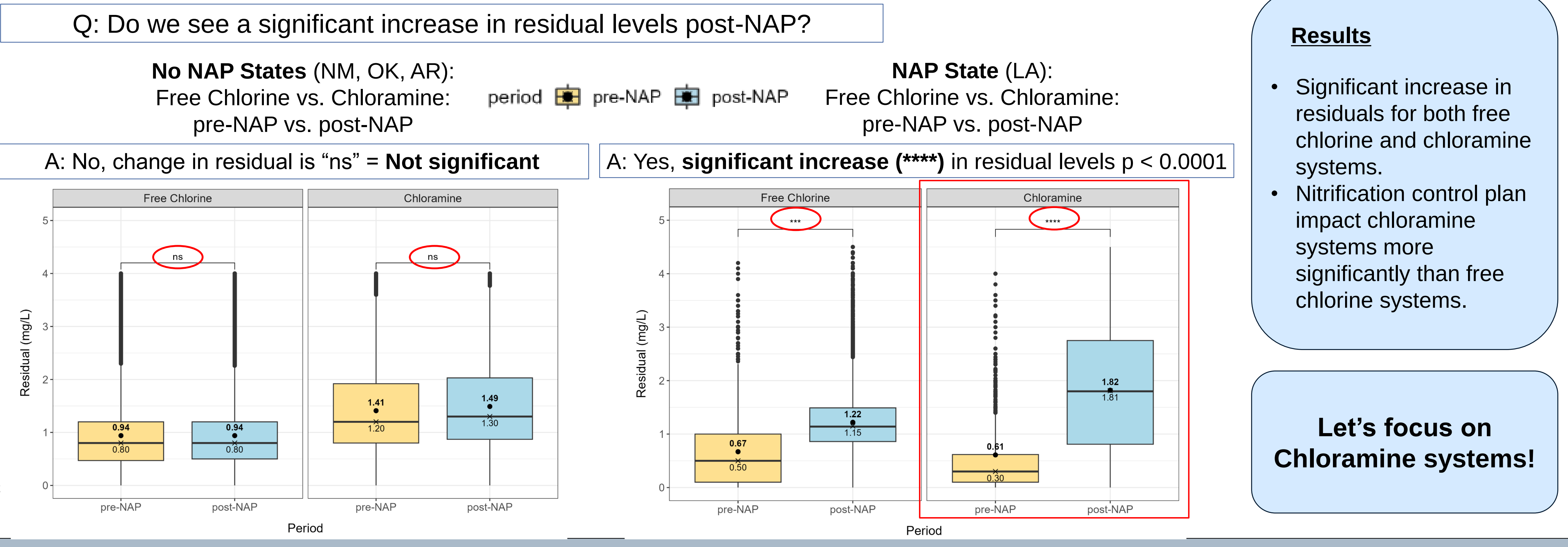
- Y_t : Outcome at "t" timepoint
- β_0 : Baseline average
- t : Time point in time
- D_t : Dummy variable indicating pre-post interventions period (pre = 0, post = 1)
- β_1 : Pre-intervention effects
- β_2 : Treatment effects
- β_3 : Effects of mean difference in the outcome between two groups
- ε_t : Error term that can be adjusted to allow for deviation from the fitted model

Results & Discussion

RO1. 6 out of 31 states (19%) that participated in survey have state mandated NAPs



Effects of NAP on Free Chlorine vs. Chloramine Systems



RO2. States with NAPs see increase in residual, reduction in TC, and no tradeoffs in MDBPs

Case studies:

Louisiana and Texas vs. non-NAP States

- Would states benefit from mandating a NAP?
- What are the potential tradeoffs of such a mandate?

Table 1. Treatment Differences
Mean (SD): Based on Texas' State NAP enforcement date (data from 2010–2020)

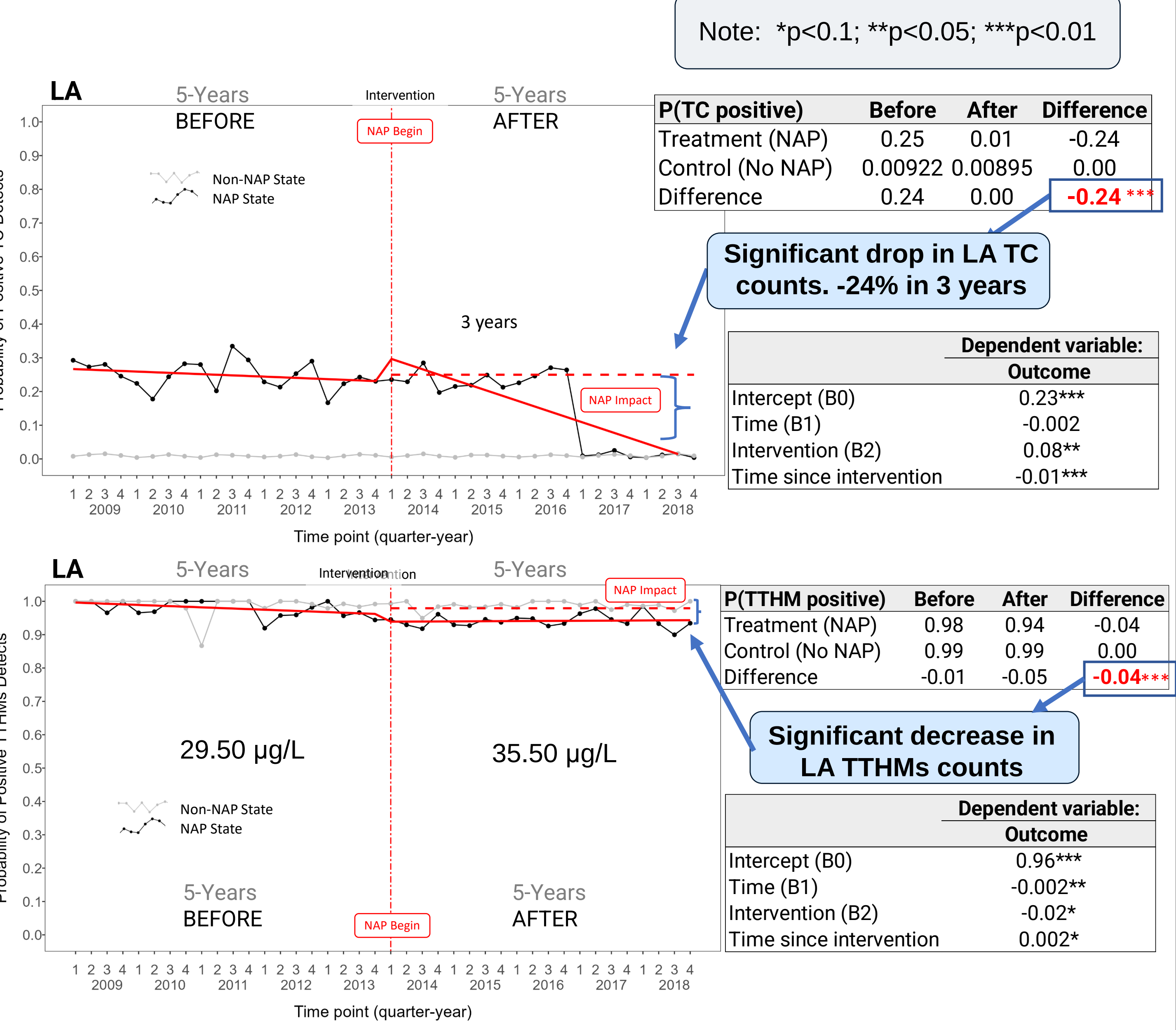
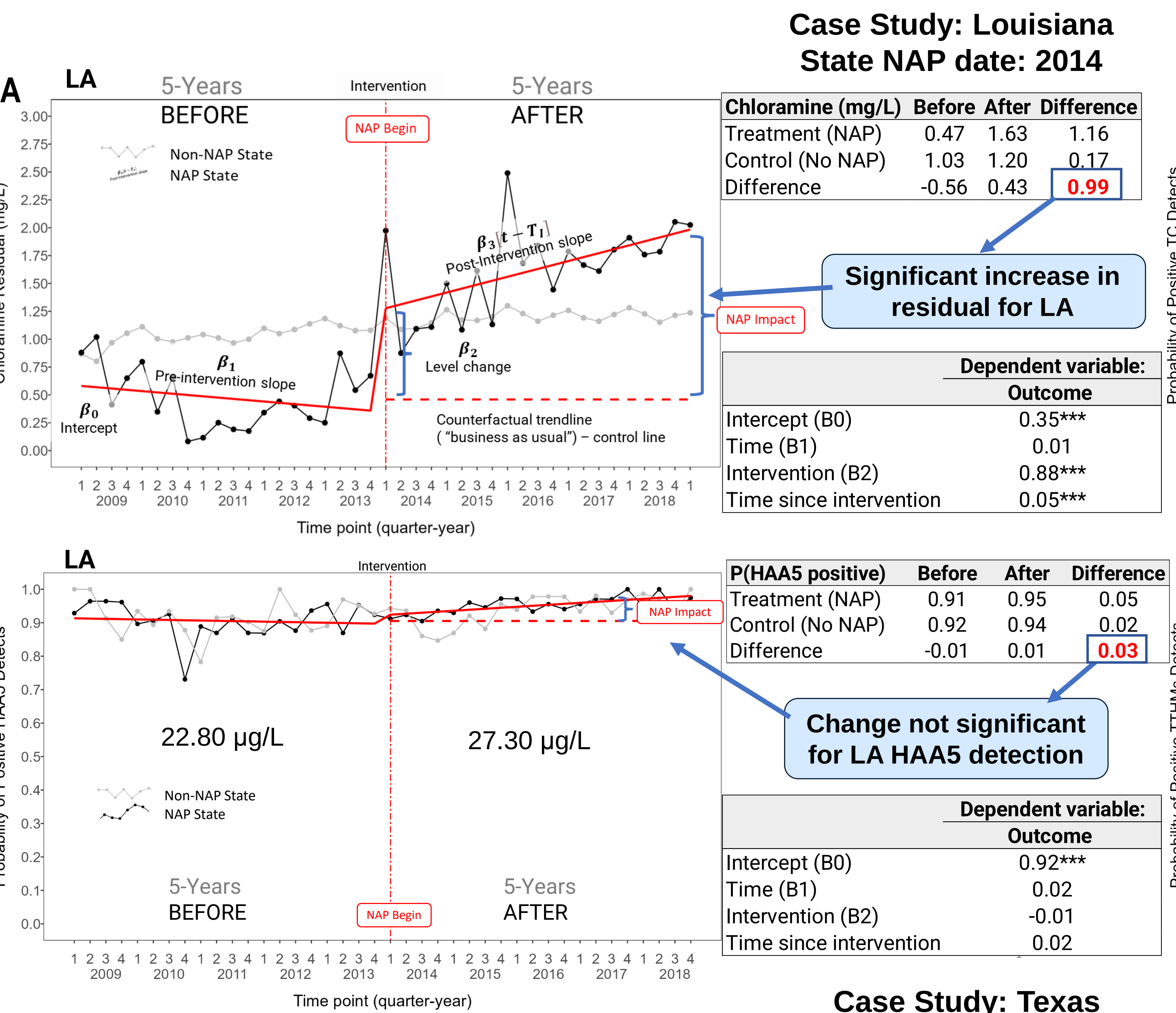
Characteristic	Non-NAP				NAP			
	pre-NAP	post-NAP	Difference ¹	95% CI ^{1,2}	pre-NAP	post-NAP	Difference ¹	95% CI ^{1,2}
Chloramine (mg/L)	1.05 (0.08)	1.20 (0.06)	0.16	0.16, 0.16	0.61 (0.27)	1.82 (0.15)	1.2	1.2, 1.2
HAA5 (µg/L)	19.70 (3.98)	28.17 (4.24)	8.5	7.9, 9.0	19.00 (2.66)	25.41 (4.33)	6.4	6.3, 6.5
TTHMs (µg/L)	44.21 (10.39)	51.33 (5.33)	7.1	5.8, 8.5	32.82 (5.33)	34.37 (4.34)	1.5	1.4, 1.7
Total coliform (% positive detects)	0.01 (0.00)	0.01 (0.00)	0	0.00, 0.00	0.28 (0.04)	0.03 (0.06)	-0.26	-0.26, -0.26
HAA5 (% positive detects)	0.86 (0.07)	0.90 (0.08)	0.04	0.04, 0.05	0.87 (0.03)	0.93 (0.04)	0.06	0.06, 0.06
TTHMs (% positive detects)	0.97 (0.02)	0.96 (0.03)	0	-0.01, 0.00	0.96 (0.03)	0.95 (0.03)	-0.01	-0.01, -0.01

¹ Welch Two Sample t-test
² CI = Confidence Interval
Voluntary implementation date: 2015; Mandatory compliance date: 2015

Table 2. Treatment Differences
Mean (SD): Based on Texas' State NAP enforcement date (data from 2009–2019)

Characteristic	Non-NAP				NAP			
	pre-NAP	post-NAP	Difference ¹	95% CI ^{1,2}	pre-NAP	post-NAP	Difference ¹	95% CI ^{1,2}
Chloramine (mg/L)	1.08 (0.06)	1.22 (0.04)	0.14	0.14, 0.14	2.06 (0.09)	2.23 (0.13)	0.17	0.17, 0.17
HAA5 (µg/L)	21.27 (4.01)	29.44 (3.68)	8.2	7.8, 8.6	18.67 (1.82)	21.68 (2.61)	3	3.0, 3.0
TTHMs (µg/L)	45.26 (9.03)	50.76 (5.42)	5.5	4.6, 6.4	34.93 (3.73)	34.21 (2.92)	-0.72	-0.78, -0.66
Total coliform (% positive detects)	0.01 (0.00)	0.01 (0.00)	0	0.00, 0.00	0.01 (0.01)	0.01 (0.00)	0	0.00, 0.00
HAA5 (% positive detects)	0.85 (0.08)	0.93 (0.05)	0.08	0.07, 0.09	0.96 (0.01)	0.97 (0.01)	0.01	0.01, 0.01
TTHMs (% positive detects)	0.96 (0.04)	0.96 (0.02)	0.01	0.00, 0.01	0.96 (0.03)	0.97 (0.01)	0.01	0.01, 0.01

¹ Welch Two Sample t-test
² CI = Confidence Interval
Voluntary implementation date: 2015; Mandatory compliance date: 2015



Conclusions & Ongoing Work

RO1. 5 OUT OF 33 STATES (15%) THAT PARTICIPATED IN SURVEY HAVE IMPLEMENTED NITRIFICATION ACTION PLANS

- Most states responded "no" to having a nitrification control plan due to having very limited utilities with chloramine systems and a lack of defined requirement in regulation.
- Utilities with state mandated nitrification control plan uses guidelines provided in industry standards such as (AWWA) Manual M56 – Nitrification, Prevention and Control in Drinking Water.

Challenges (from states' perspectives in the implementation of the Nitrification Action Plans): "NAP is not a one size fits all and when distribution condition changes NAP needs to be updated, getting operators to understand the chlorine curve, lack of training & experience by state staff and PWS operators."

RO2. STATES WITH NITRIFICATION ACTION/MANAGEMENT/CONTROL PLANS SEE INCREASE IN DISINFECTANT RESIDUAL, REDUCTION IN TOTAL COLIFORM COUNTS, AND NO SIGNIFICANT INCREASE IN DISINFECTION BYPRODUCTS

- Louisiana (residual started low) – NAP impact is significant.
- Texas (residual started high) – NAP impact is not significant.
- Effective in increasing disinfection residuals for states with low residuals, reducing total coliform counts, does not significantly increase DBPs.
- The tradeoff between high residual disinfectant and disinfection byproducts (DBPs) is an important consideration in water treatment. Here we did not see high tradeoff. –GOOD!

FUTURE WORKS: RO3: Get more states to share state database and evaluate Nitrate-Nitrite, pH, Temperature, MDBP data as well as study ways to overcome challenges addressed by states.

Acknowledgements:

I would like to express my deepest gratitude to my PI, Dr. Katherine Alfredo (USF), for her guidance and invaluable feedback. I also could not have undertaken this project without the support of Alan Roberson and Stephanie Schlea (ASDWA), and State committee members who contributed to this project and for their willingness to share state data and provided valuable survey responses to make this analysis possible. Lastly, I would be remiss in not mentioning my family, and my colleagues Andrew Black, Horace Jakpa, and Monica Castro Carrias for great feedback sessions and moral support.

Contact: Natchaya Luangphairin natchayal@usf.edu (812) 318-2456, Katherine Alfredo kalfredo@usf.edu, Alan Roberson aroberson@asdwa.org, Stephanie Schlea sschlea@asdwa.org